

Bubble sort

Output of sorting in programming ?

- Make array/data set $A = [x_1, x_2, \dots, x_n]$ satisfy this condition for each index:
 - $A[i] \leq A[i+1]$ always return true for ASC order
 - $A[i] \geq A[i+1]$ always return true for DESC order

Main idea of bubble sort ?

- Through each iteration, compare current variable with value next to it
 - If $x[i] > x[i+1]$, swap 2 position
 - if $x[i] < x[i+1]$, do nothing, already sorted
 - By doing it, we can always move the largest value to the end position
- For the next loop, because the last i positions is sorted, we only do it for $N-i$ part
- The largest unsorted element is going up to the end in each pass, that's why it called bubble sort (like a bubble)
- Keep doing it! We end up with $1 + 2 + \dots + N$ times, we have formula: $N(N+1)/2$

Time complexity ?

- Since we do $N(N+1)/2$ operation = $N^2 + N/2$
- Drop all constant and insignificant values $\rightarrow O(N^2)$